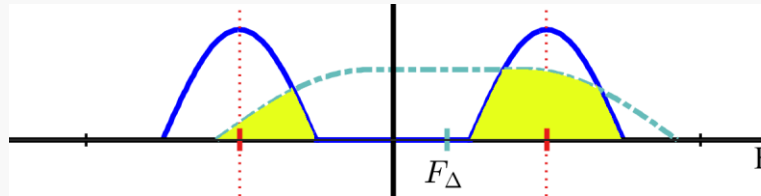


Carrier Frequency Synchronization

Qasim Chaudhari



Introduction

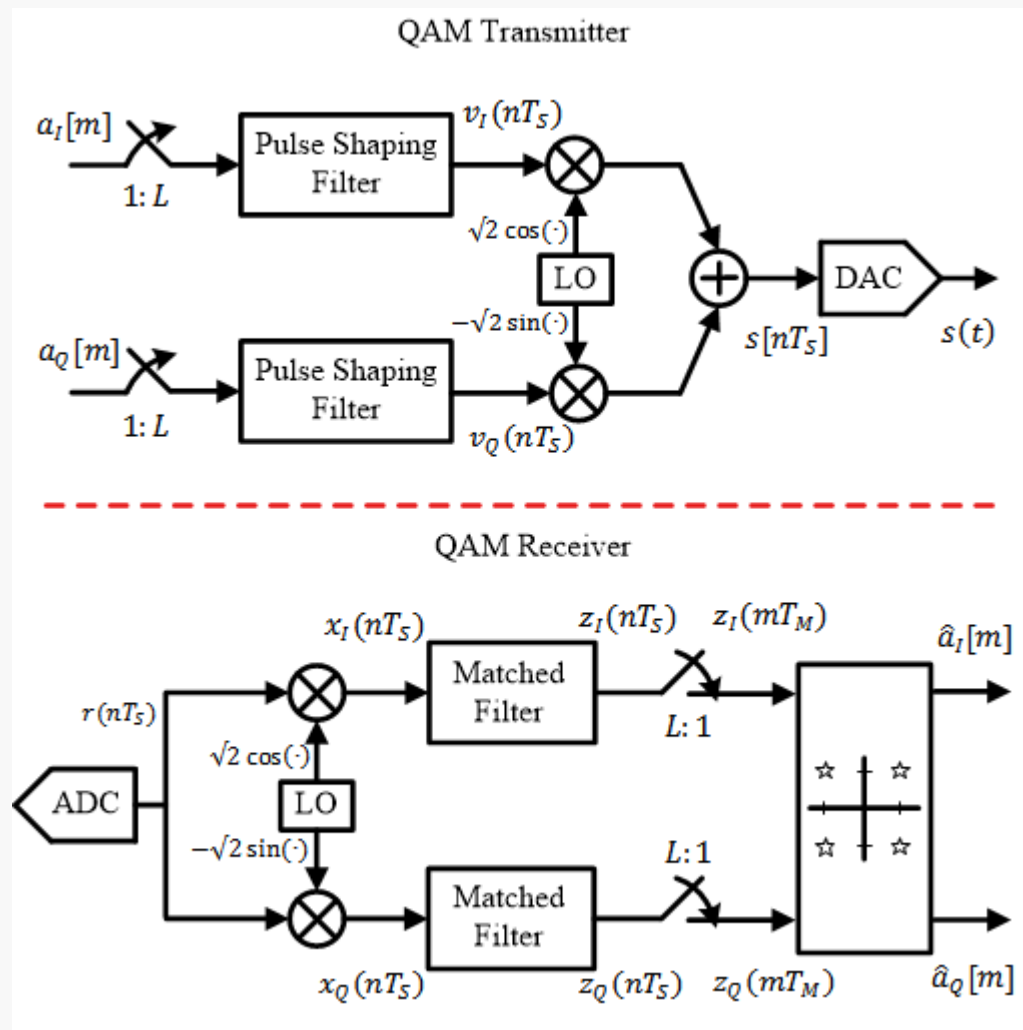
For QAM and PSK, the Rx LO is not phase and frequency aligned with the carrier of the incoming signal

A CFO usually arises due to two reasons

- Frequency mismatch between the Tx and Rx oscillators
- Doppler effect

A QAM Transceiver

- Symbol rate $1/T_M$
- Rx sample rate $1/T_S$
- L samples/symbol, $T_S = T_M/L$
- Matched filter output $z(mT_M)$



ppm Rating

ppm rating of the oscillator crystal indicates how much its frequency may deviate from the nominal value

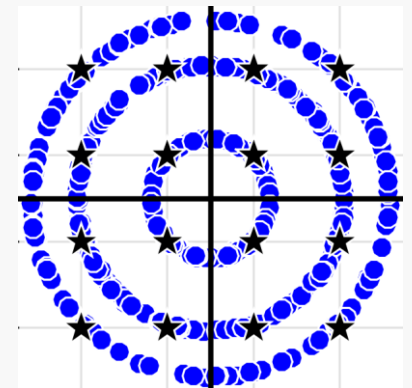
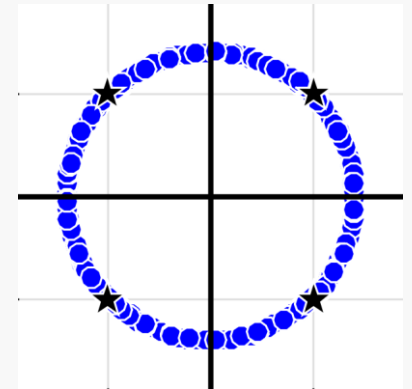
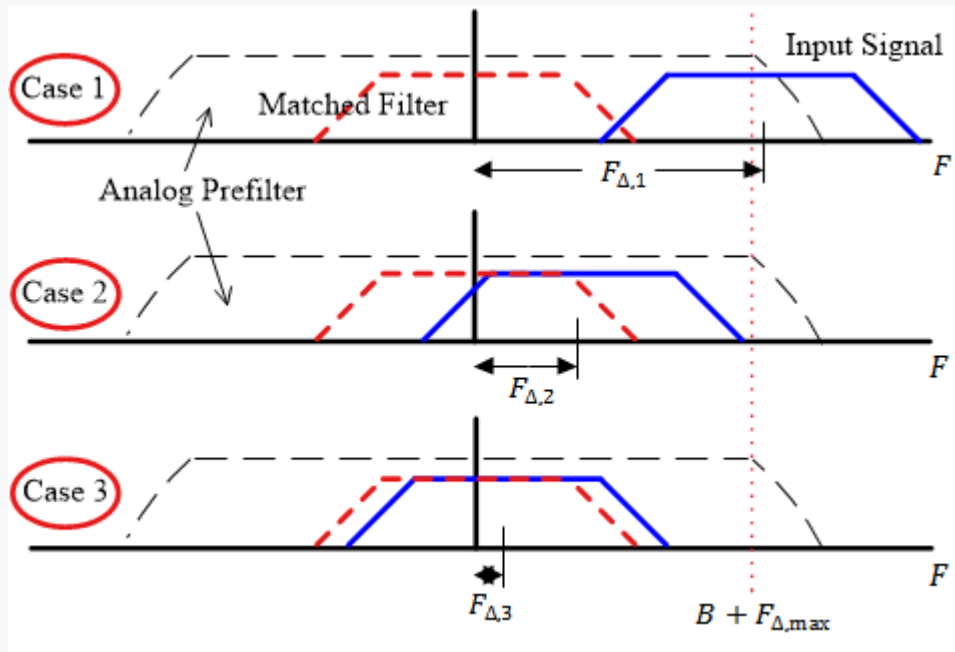
- 1 ppm is 1 out of 10^6 parts, just like a percent is 1 part in hundred

Consider an example of a wireless system operating at 2.4 GHz and with ± 20 ppm crystals

- Maximum frequency deviation becomes

$$\pm \frac{20}{10^6} \times 2.4 \times 10^9 = \pm 48 \text{kHz}$$

3 Scenarios



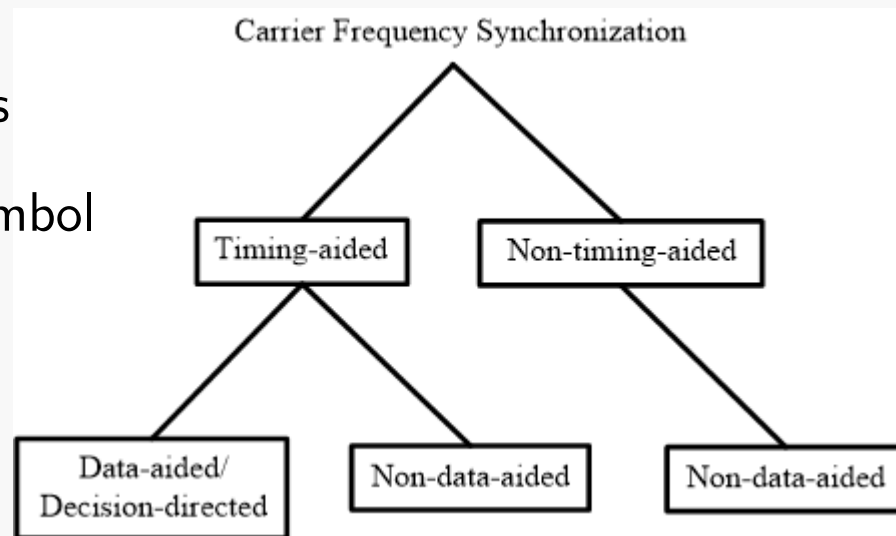
Case 3: A small CFO

- no change in magnitude
- phase keeps changing thus rotating the constellation

The Big Picture

Non-timing-aided techniques

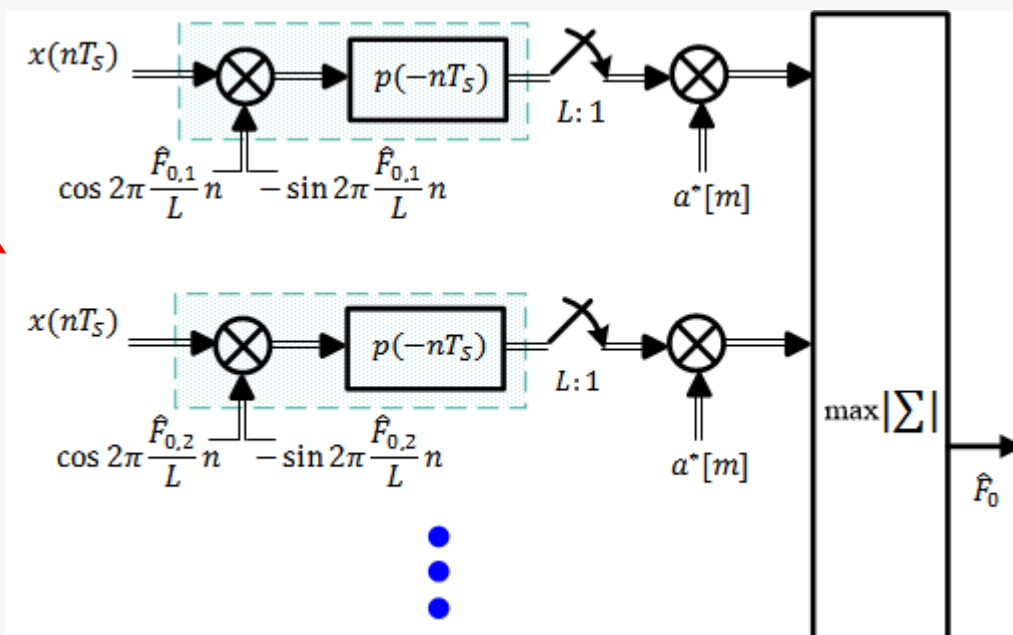
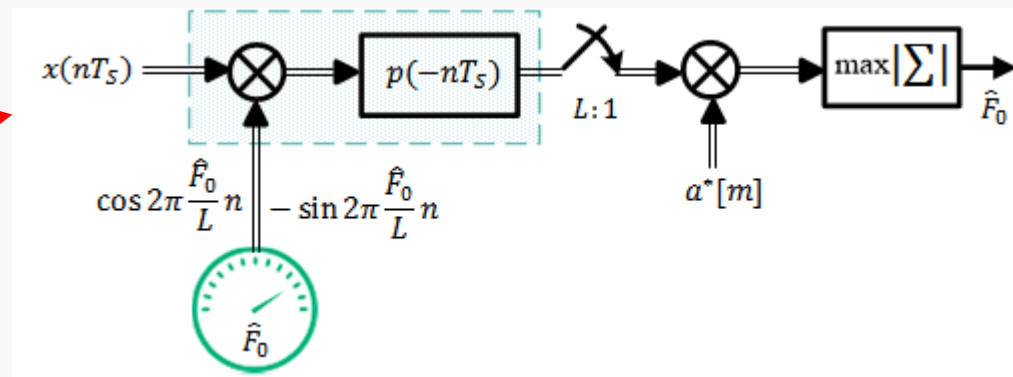
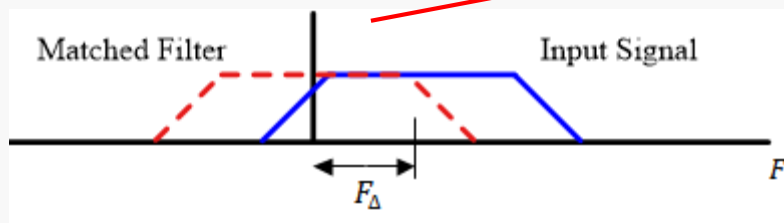
- Not enough overlap with Rx filters
- No downsampling to 1 sample/symbol
- Inherently non-data-aided



Timing-aided techniques

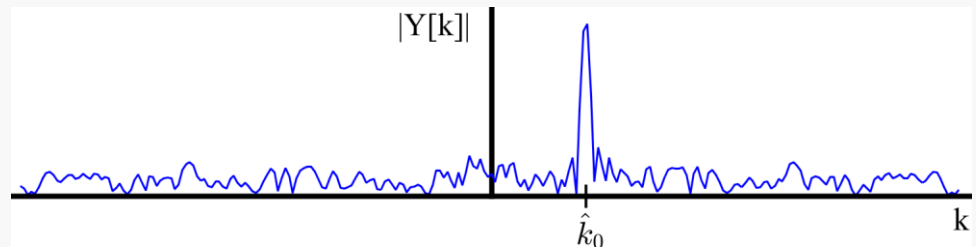
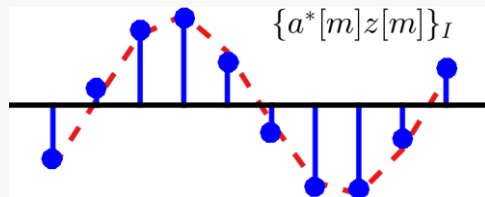
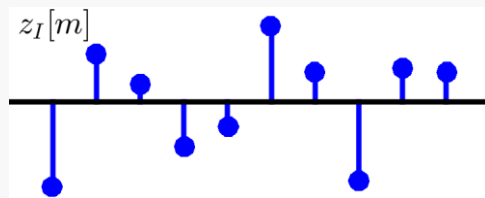
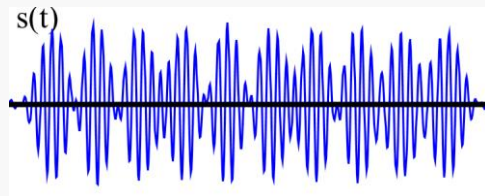
- Timing can be recovered from slowly rotating constellation
- Downsampled to 1 sample/symbol
- Both data-aided and non-data-aided techniques possible

Brute Force Estimator



The DFT Estimator

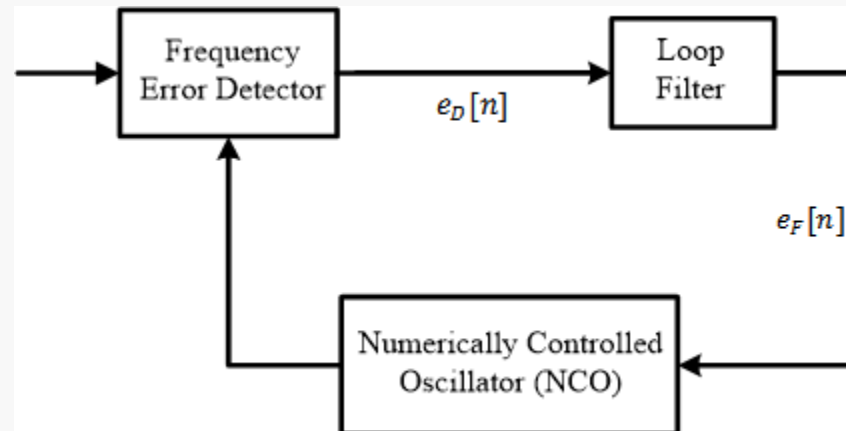
Problem of finding the CFO F_0 in a modulated signal can be reduced to estimating the frequency F_0 of a simple complex sinusoid embedded in noise



$$\hat{k}_0 \Rightarrow \max |Y[k]|$$

Frequency Locked Loop (FLL)

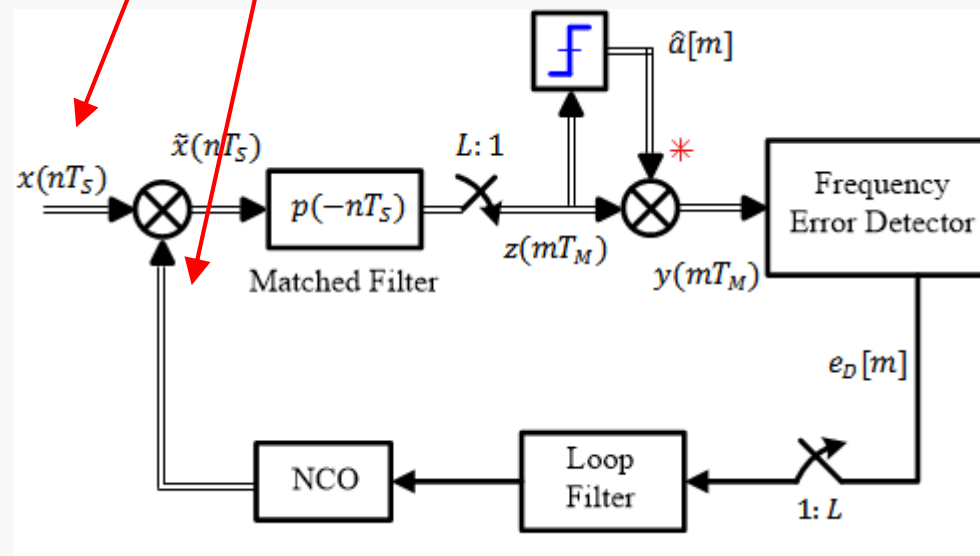
- Why is it needed?
- Just like a PLL, an FLL consists of three components



Phase Based FED

FED = Frequency Error Detector

$$F_{0:e} = F_0 - \hat{F}_0$$

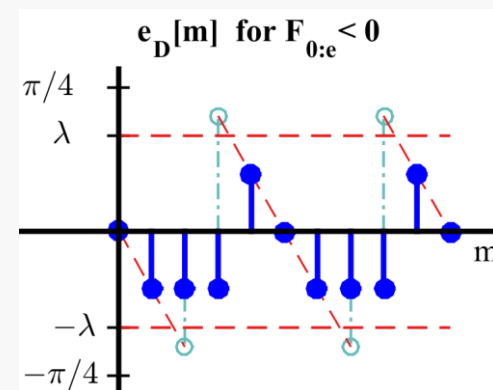
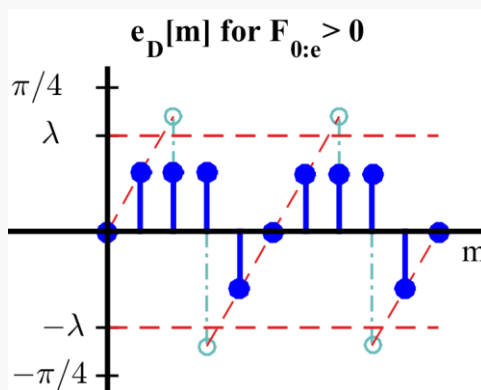
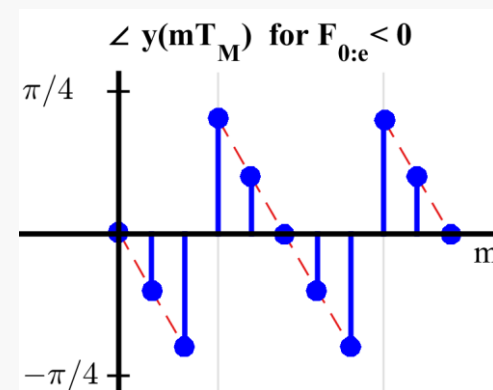
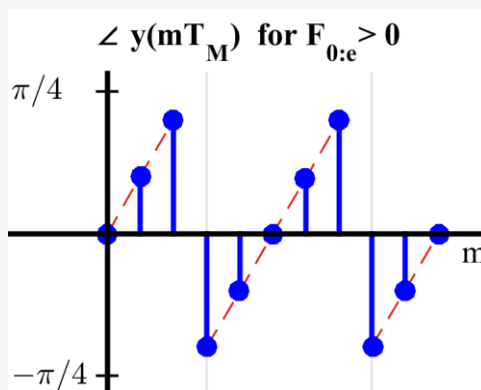


Phase Based FED

Guided by correlation, we form $y(mT_M) = \hat{a}^*[m] \cdot z(mT_M)$

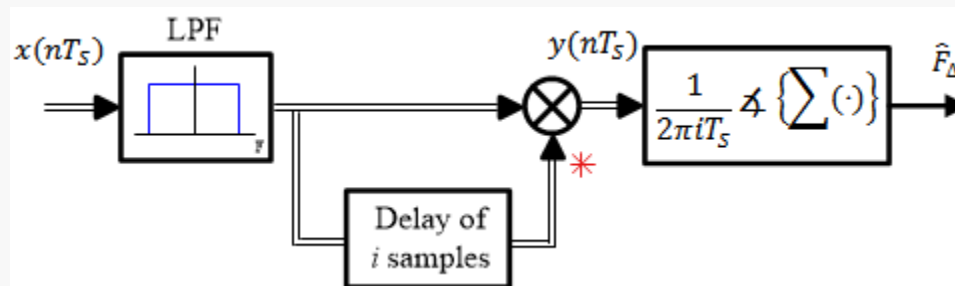
Error signal

$$e_D[m] = \begin{cases} \angle y(mT_M) & -\lambda < \angle y(mT_M) < \lambda \\ e_D[m-1] & \text{otherwise} \end{cases}$$



Delay and Multiply Method

- Definition of phase $\Delta\phi = 2\pi F(T_2 - T_1)$
- **Main idea:** Frequency of any complex sinusoid can be estimated by taking a phase difference between any of its two samples at known time instants, say T_2 and T_1
- The frequency can be found from
$$F = \frac{\Delta\phi}{2\pi(T_2 - T_1)}$$



Derivative FED

- Correlation lead us to remove the modulation

$$\text{corr}[0] = \sum_{m=0}^{N_0-1} a^*[m]z(mT_M)$$

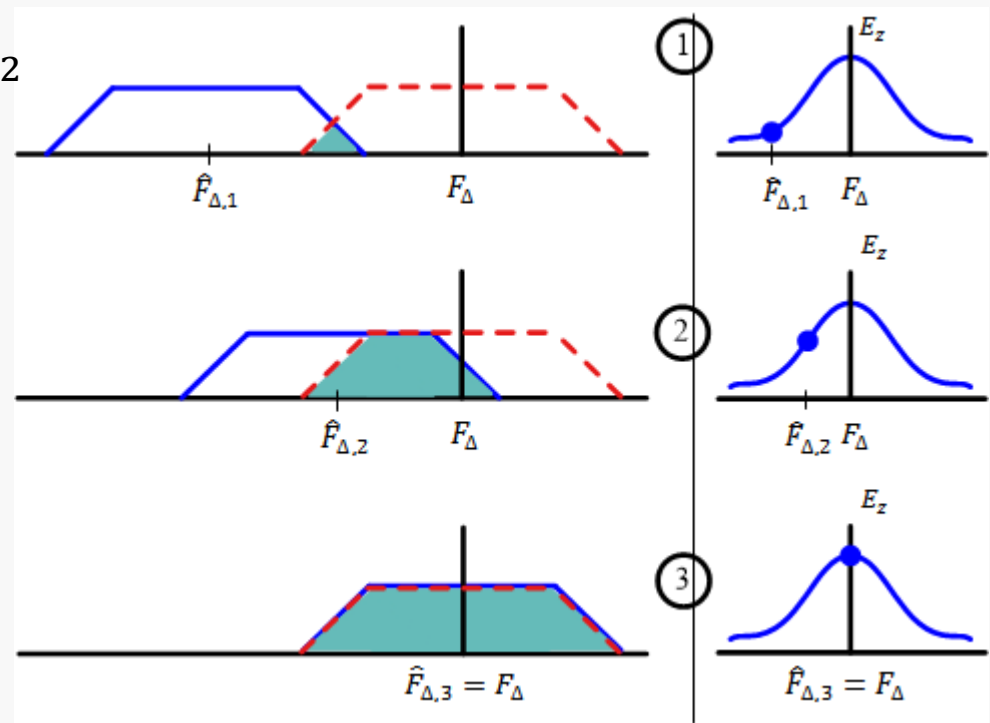
Modulated symbol

Matched filter output

- Main idea:** Remove the modulation by multiplication with a value similar in sign, so we replace $a[m]$ with the matched filter output $z(nT_S)$
- This leads to the correlation output as $\sum_n |z(nT_S)|^2$ which is the input signal energy

Derivative FED

- Why it works?
- Target of a non-timing-aided CFO compensation technique is to maximize this energy
- Feedback $\rightarrow$ we focus on $|z(nT_S)|^2$

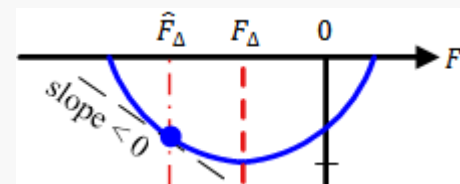


Derivative FED

- To find the maximum of the curve, we equate the derivative to zero
- Purpose of an FLL is to drive an error signal to zero as well
- An error signal can be formed as

$$e_D[n] = \left\{ z(nT_S) \cdot \left\{ \dot{z}(nT_S) \right\}^* \right\}_I$$

- Roles of $z(nT_S)$ and $\dot{z}(nT_S)$

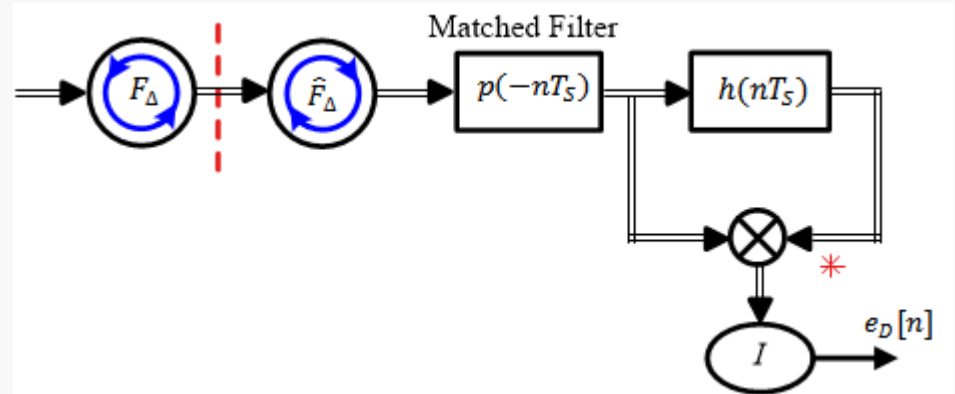


Derivative FED

Two possible routes

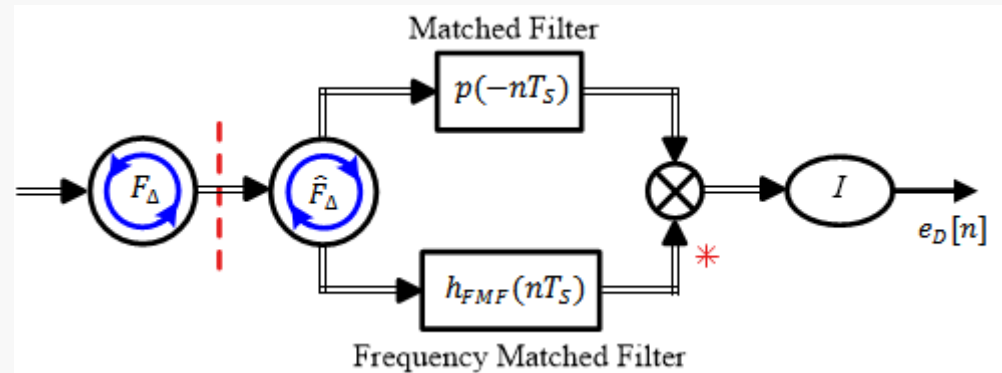
Serial implementation using a derivative filter

$$e_D[n] = \left\{ z(nT_S) \cdot \left\{ \dot{z}(nT_S) \right\}^* \right\}_I$$



Parallel implementation using a
Frequency Matched Filter (FMF)

$$e_D[n] = \left\{ z(nT_S) \cdot z_{FMF}^*(nT_S) \right\}_I$$



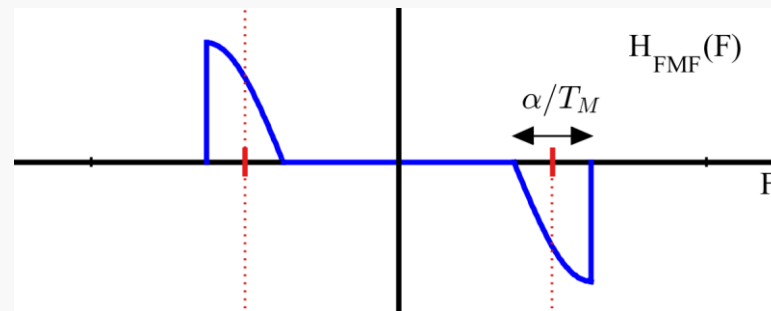
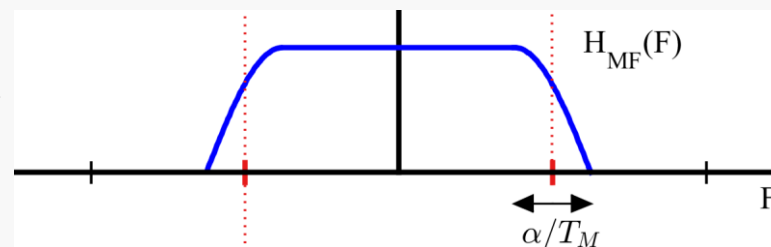
Frequency Matched Filter (FMF)

It is easy to understand why the FMF is also called a band edge filter

Only transition bandwidth contributes towards forming the FMF

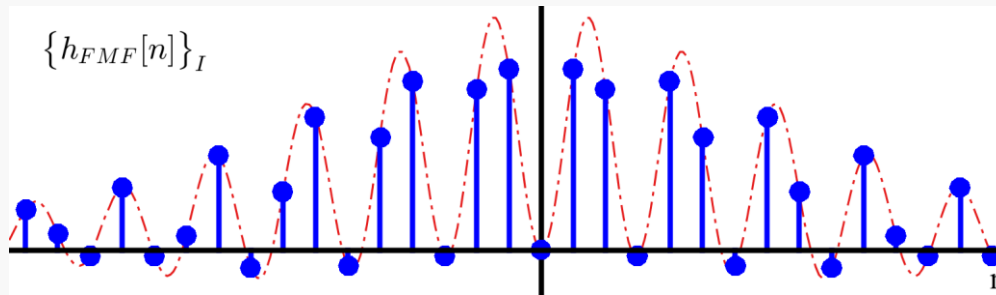
Excess bandwidth provides the energy for
synchronization purpose!

fred harris



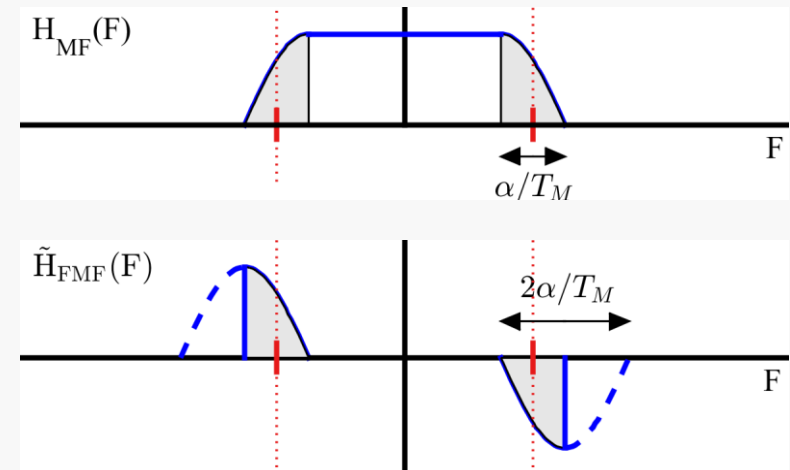
Modified FMF

Drawback of FMF: discontinuity at the edges



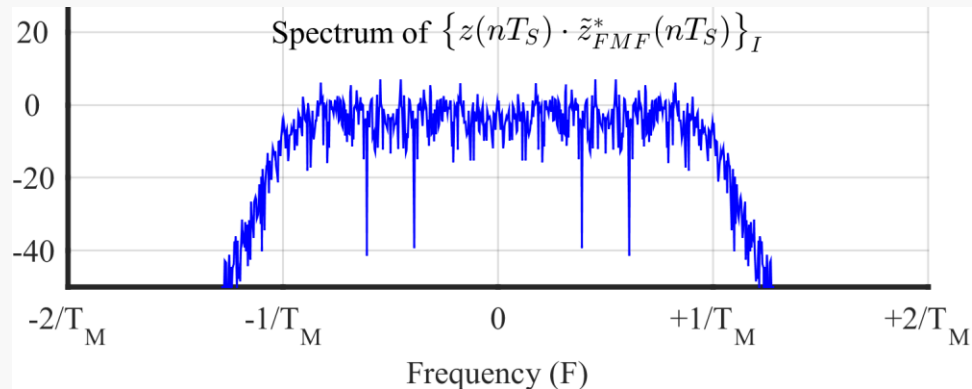
One solution: continue the spectral shape past the discontinuity in the same trajectory

The final spectral transition forms a half cycle of a sine wave

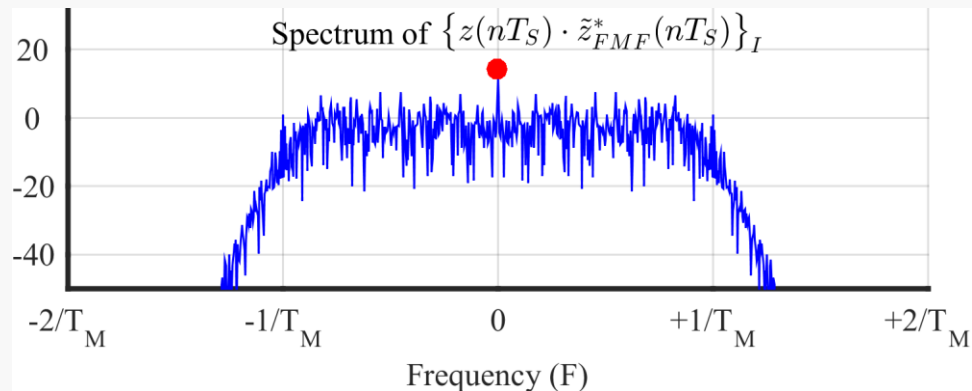


Modified FMF

Convolution in frequency domain between the spectrum of the matched filter output (full signal bandwidth) and that of the modified FMF output (limited in bandwidth)



No CFO

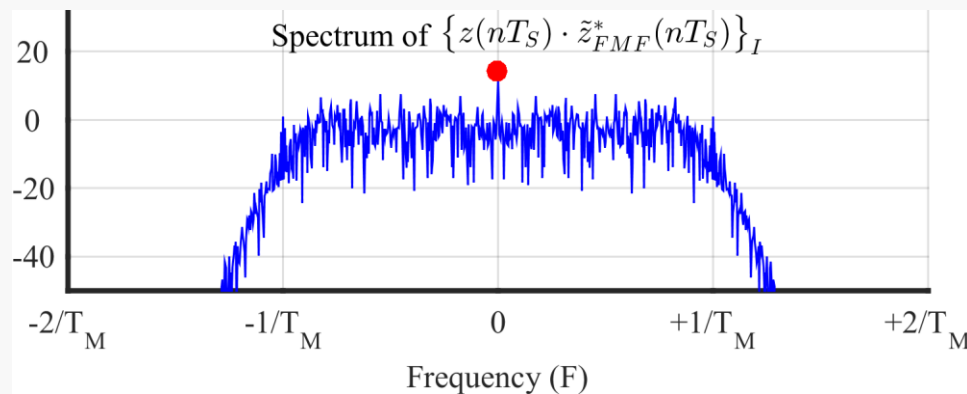


With CFO

Modified FMF

Main point: the problem here is with the full bandwidth matched filter

Since the signal at these frequencies arises due to interactions among data symbols themselves, it is often termed as **self noise (modulation noise or pattern noise)**



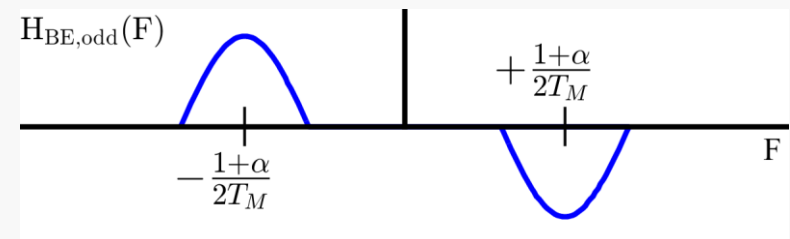
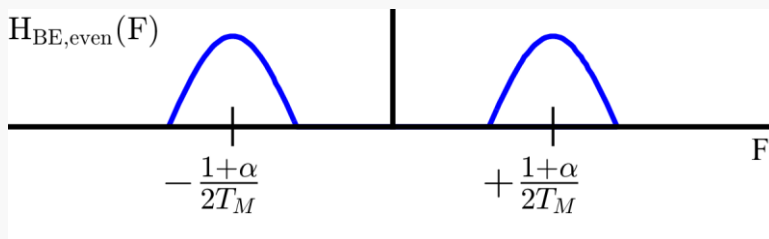
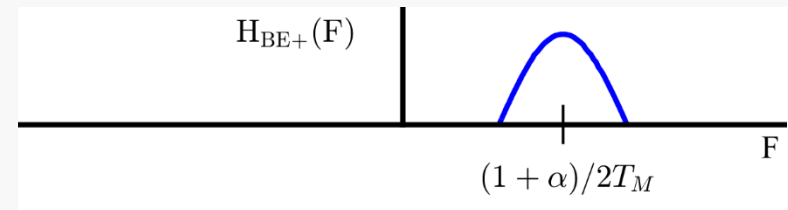
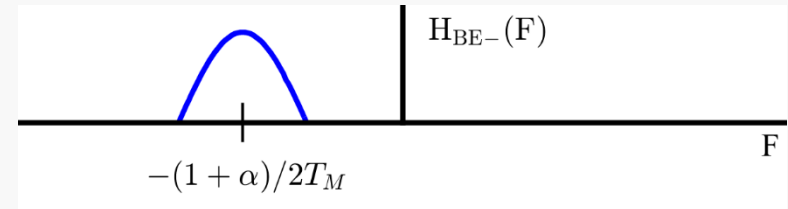
Band Edge FLL

Break the modified FMF into two parts

- An upper band edge filter $H_{BE+}(F)$
- A lower band edge filter $H_{BE-}(F)$

We obtain two filters

- Even band edge filter through addition
- Odd band edge filter through subtraction

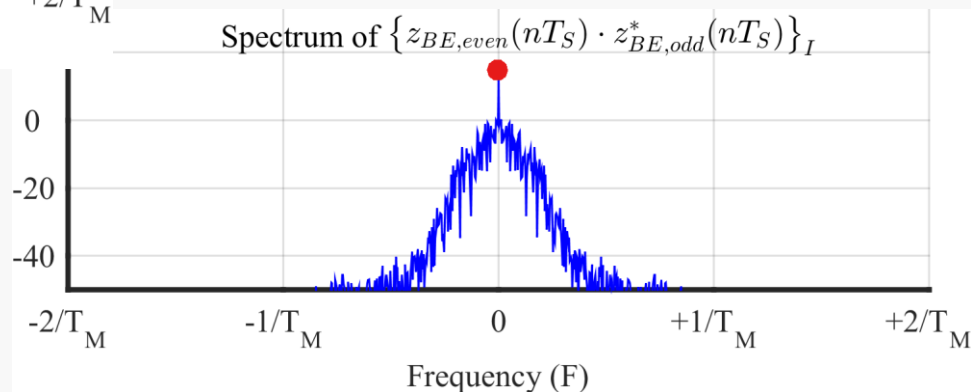
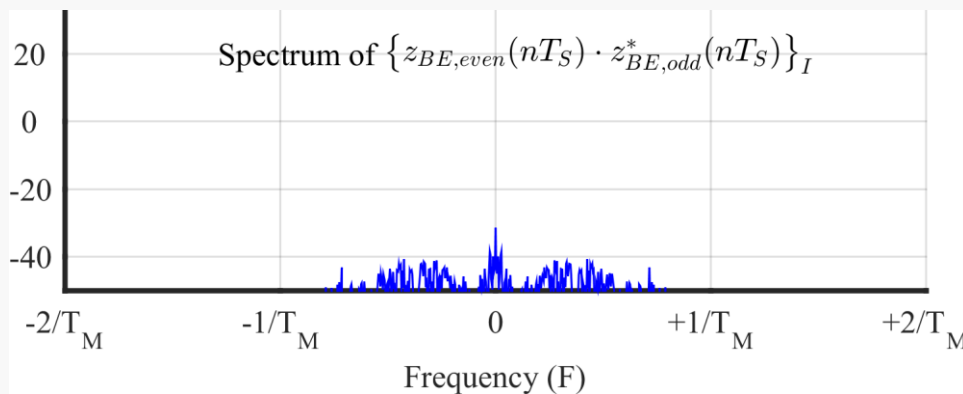


Band Edge FLL

Replace the matched filter output with the even band edge filter output

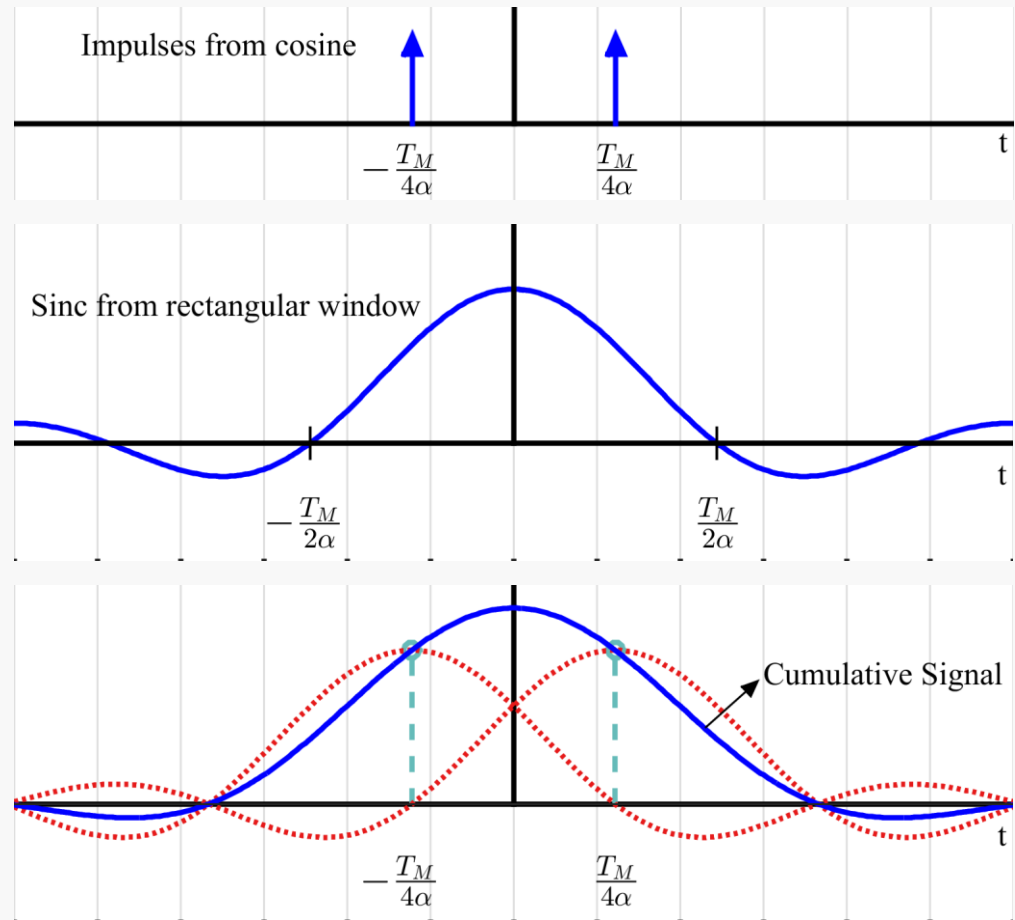
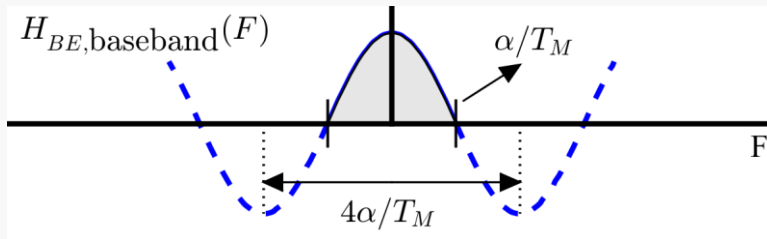
Form the error signal as

$$e_D[n] = \left\{ z_{BE,even}(nT_S) \cdot z_{BE,odd}^*(nT_S) \right\}_I$$



Band Edge FLL

- For impulse response, consider the baseband spectral lobe that is a simple half-cosine



Band Edge FLL: Intuition

- Recall the error signal

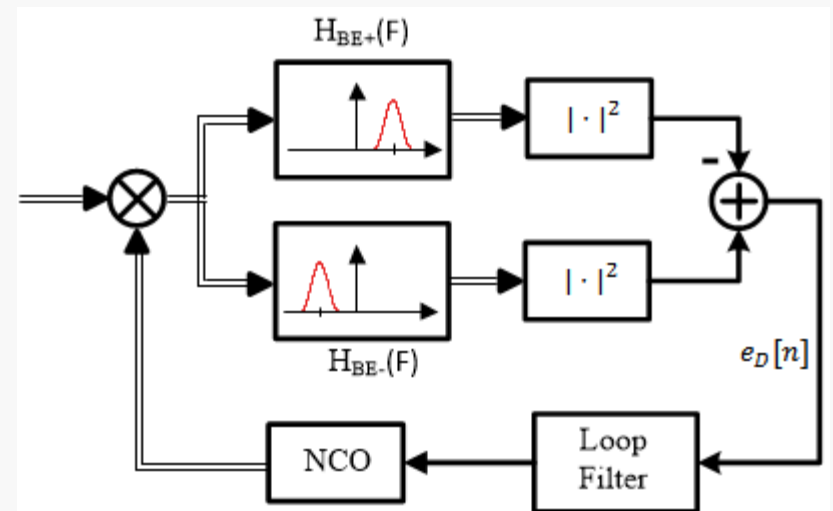
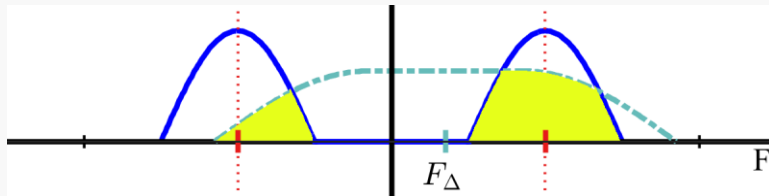
$$e_D[n] = \left\{ z_{\text{BE,even}}(nT_S) \cdot z_{\text{BE,odd}}^*(nT_S) \right\}_I$$

- This can be simplified as

$$e_D[n] = |z_{\text{BE-}}(nT_S)|^2 - |z_{\text{BE+}}(nT_S)|^2$$

- A band edge FED is just an energy difference between the outputs of lower and upper band edge filters!
 - Also known as Differential Power Measurement (DPM) FED

Band Edge FLL: Intuition



GNU Radio Example

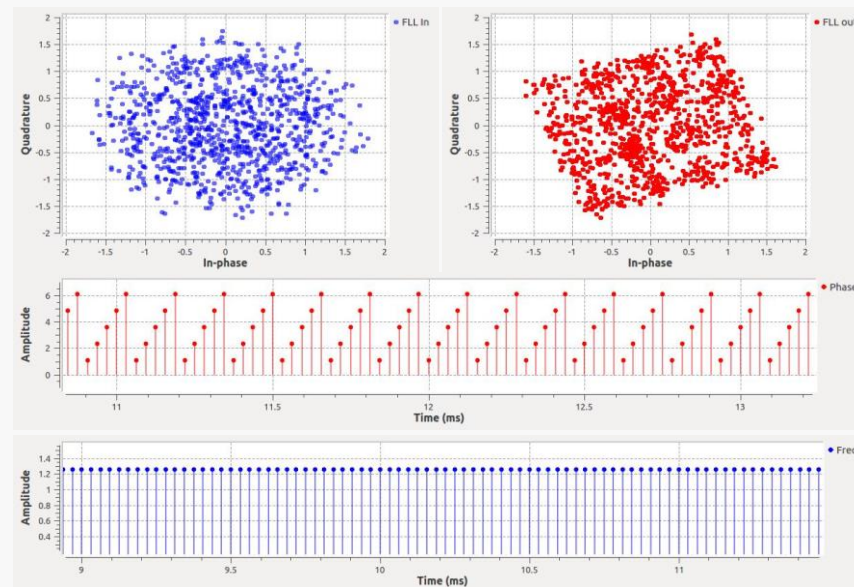
FLL Band Edge block

- Samples per symbol L
- Filter rolloff factor α
- Prototype filter size
- Loop bandwidth

Example: 16-QAM, CFO -0.2,

loop bandwidth $2\pi/250$,

$\alpha = 0.5$, $L = 4$



Small Picture

Small CFO

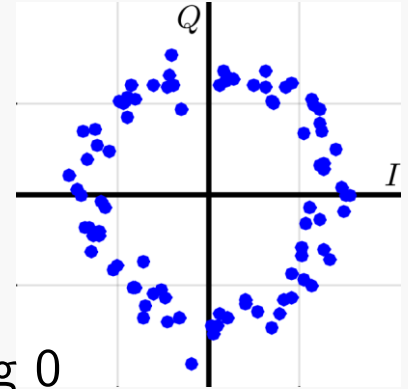
- Matched filtered signal is downsampled to 1 sample/symbol
- Modulation is removed through data symbol $a^*[m]$
- CFO is the frequency of the resultant complex sinusoid

Large CFO

- Energy maximization perspective of matched filter output

Multiple Correlations Estimator

- At low SNR, the signal is easily overwhelmed by noise



- Main idea:** use multiple correlations instead of just one at lag 0

- For example

$$\text{corr}[1] = \frac{1}{N_p - 1} \sum_{m=1}^{N_p-1} y(mT_M) y^*[(m-1)T_M]$$

- Extending the same concept by taking the auto-correlation at lag j

$$\text{corr}[j] = \frac{1}{N_p - j} \sum_{m=j}^{N_p-1} y(mT_M) y^*[(m-j)T_M]$$

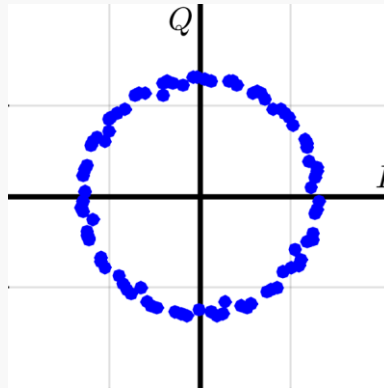
Multiple Correlations Estimator

- This turns out to be

$$\text{corr}_I[j] = \cos 2\pi F_0 j$$

$$\text{corr}_Q[j] = \sin 2\pi F_0 j$$

- It is much smoother because the averaging operation as a filter reasonably suppressed the noise



- Observe that

$$\angle \text{corr}[j] = 2\pi F_0 j$$

Multiple Correlations Estimator

Fitz method

- Averaging both sides of $\angle \text{corr}[j]$,

$$\frac{1}{N_F} \sum_{j=1}^{N_F} \angle \text{corr}[j] = \frac{1}{N_F} \sum_{j=1}^{N_F} 2\pi F_0 j = \pi F_0 (N_F + 1)$$

$$\hat{F}_0 = \frac{1}{\pi N_F (N_F + 1)} \sum_{j=1}^{N_F} \angle \text{corr}[j]$$

LR method

- Take average first and angle later

$$\hat{F}_0 = \frac{1}{\pi (N_{\text{LR}} + 1)} \angle \left\{ \sum_{j=1}^{N_{\text{LR}}} \text{corr}[j] \right\}$$